Chapter 5 — Thermal Design, Power Management, and TCO

> *"Every watt you waste cooling hardware is a watt you didn't spend training a model. The most important equation in AI infrastructure isn't a matrix multiply — it's Power × PUE × $/kWh × 8,760 hours."* — Principal Infrastructure Architect's Cost Axiom

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Chapter Overview

The H100 SXM5 has a Thermal Design Power (TDP) of 700W. A rack of 8 DGX H100 nodes draws 120+ kilowatts. A 10,000-GPU training cluster consumes power comparable to a small town. At this scale, thermal management is not a facilities concern — it is a performance engineering concern. Thermal throttling directly reduces MFU. Poor power allocation inflates TCO by millions of dollars per year.

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- 5.1 Where Heat Comes From — Semiconductor Power Dissipation
- 5.2 GPU Thermal Architecture and the Throttle Mechanism
- 5.3 Cooling Technologies — Air, DLC, and Immersion
- 5.4 GPU Power Management — DVFS and Power States
- 5.5 Power Budgeting — From Chip to Rack to Datacenter
- 5.6 TCO Modeling for AI Infrastructure
- 5.7 Performance-Per-Watt — The True Optimization Target
- 5.8 Build vs. Buy vs. Rent — The Decision Framework
- 5.9 Thermal Monitoring and Remediation
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5.1 Where Heat Comes From — Semiconductor Power Dissipation

Before managing heat, you must understand what generates it. All GPU power dissipation traces back to three sources in digital CMOS circuits.

5.1.1 The Three Sources of Power Dissipation

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POWER DISSIPATION IN CMOS CIRCUITS:
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  P_total = P_dynamic + P_static + P_short_circuit

1. DYNAMIC POWER (dominant — 60-75% of total):
   P_dynamic = alpha × C × V² × f

   where:
     alpha = activity factor (0-1, fraction of transistors switching per cycle)
     C     = total switching capacitance
     V     = supply voltage
     f     = clock frequency

   KEY INSIGHT: Power scales as V².
   Reducing voltage by 10% cuts dynamic power by ~19%.
   This is the fundamental lever behind DVFS.

   Activity factor for AI workloads:
     Dense GEMM (all SMs active):    alpha ≈ 0.40-0.55 (40-55% of transistors
switch)
     Decode (memory-bound):          alpha ≈ 0.15-0.25 (ALUs mostly idle, memory
active)
     Idle / low utilization:         alpha ≈ 0.02-0.05 (mostly clock gating)

2. STATIC POWER / LEAKAGE (growing per generation — 20-35%):
   P_static = V_dd × I_leakage

   As transistors shrink (5nm → 3nm → 2nm), gate oxide becomes thinner,
   increasing leakage current.
   At 4nm (H100): leakage consumes ~25% of total power even at zero workload.
   This is why idle H100 GPUs still draw 100-150W.

   Implication: GPU idle power is not free.
   A 700W GPU cluster idling at 15% draws 105W × N_GPUs — still significant.

3. SHORT-CIRCUIT POWER (<5%):
   Briefly conducted when both PMOS and NMOS transistors switch simultaneously.
   Minimized by circuit design. Not a system-level concern.

THE DVFS SWEET SPOT:
  The V² relationship means voltage reduction is non-linear:
    10% voltage reduction → 19% dynamic power reduction → 10% frequency reduction
    15% voltage reduction → 28% dynamic power reduction → 15% frequency reduction
    20% voltage reduction → 36% dynamic power reduction → 20% frequency reduction

  At 700W TDP with 70% dynamic power:
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    Reducing V by 10%: saves 700 × 0.70 × 0.19 = 93W
    Frequency cost:    loses 10% of compute throughput

  The last 10% of frequency (boost clock) costs disproportionate power.
  This is why the 90-95% TDP operating point is the efficiency optimum.
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5.1.2 Where Power Goes Inside an H100 SXM5

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H100 SXM5 POWER BUDGET BREAKDOWN (700W total TDP): [ESTIMATED]

  Component                     Power Draw      % of TDP
  ──────────────────────────────────────────────────────
  Tensor Cores (GEMM)           ~200-280W       29-40%
  CUDA cores (elementwise)      ~80-120W        11-17%
  HBM3 memory subsystem         ~40-60W         6-9%
  L2 cache and SMEM             ~30-50W         4-7%
  NVLink 4.0 PHY                ~25-40W         4-6%
  Clock distribution             ~20-30W         3-4%
  Leakage (static)              ~140-175W       20-25%
  I/O, sensors, PMU             ~30-50W         4-7%
  ──────────────────────────────────────────────────────
  TOTAL                         ~700W           100%

  WORKLOAD-DEPENDENT VARIATION:
    Dense training GEMM:         ~650-700W  (compute-bound, high activity)
    Decode inference (batch=1):  ~350-450W  (memory-bound, SMs often idle)
    Decode inference (batch=64): ~500-600W  (more SMs active, higher activity)
    Idle:                        ~100-150W  (leakage + clock + fabric only)

  PRACTICAL IMPLICATION:
    A cluster running batch=1 decode (memory-bound, sparse compute) may draw
    only 400W per GPU — but the infrastructure was sized for 700W.
    Power-aware workload scheduling can pack more jobs per rack if decode
    and training jobs are mixed (since decode rarely peaks the power budget).
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5.2 GPU Thermal Architecture and the Throttle Mechanism

Understanding how a GPU responds to heat is essential for diagnosing performance variability in production clusters.

5.2.1 The Thermal Resistance Chain

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HEAT FLOW PATH — H100 SXM5 (Air Cooled DGX H100):

  GPU Junction (T_j max = 83°C)
      |
      | R_die = ~0.05°C/W  (silicon die to integrated heat spreader)
      v
  Integrated Heat Spreader (IHS)
      |
      | R_TIM = ~0.10°C/W  (thermal interface material — critical path)
      v
  Cold Plate or Heatsink Base
      |
      | R_heatsink = ~0.15°C/W (DLC) to ~0.40°C/W (air)
      v
  Coolant (water or air)
      |
      | R_fluid = variable (ambient conditions)
      v
  Ambient Environment (T_ambient ≈ 35°C in hot aisles)

THERMAL BUDGET CALCULATION:

  Air cooling (total R_thermal ≈ 0.07°C/W):
    ΔT at 700W = 700W × 0.07°C/W = 49°C rise above ambient
    Max ambient = 83°C - 49°C = 34°C
    -> Hot-aisle temperature must stay ≤ 34°C — tight headroom!

  DLC (total R_thermal ≈ 0.03°C/W):
    ΔT at 700W = 700W × 0.03°C/W = 21°C rise above coolant
    Coolant inlet ≤ 62°C allowed  -> abundant headroom, stable operation

  DLC at B200 (1,000W):
    ΔT = 1000W × 0.03°C/W = 30°C rise above coolant
    Coolant inlet ≤ 53°C — manageable with standard chilled water
    Air cooling at 1,000W: ΔT = 1000 × 0.07 = 70°C rise
    Required ambient ≤ 83 - 70 = 13°C — impossible in any real datacenter
    -> B200 REQUIRES DLC. Air cooling is physically untenable.
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5.2.2 The Thermal Throttle Mechanism — How Heat Becomes MFU Loss

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H100 THERMAL THROTTLE BEHAVIOR:
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  Temperature    State             Performance Impact
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  T_j < 75°C    P0: Max boost     Full boost clock, 0% throttle
  75 ≤ T_j < 80 P0: Soft throttle Boost clock begins stepping down
                                  ~1-3% frequency reduction per °C
  80 ≤ T_j < 83 P0/P1: Hard       Frequency reduced 5-15%
                throttle warning  DCGM throttle reason set
  T_j = 83°C    Hard throttle     Clock capped to base frequency
                engaged           ~15-30% throughput loss
  T_j > 83°C    Emergency         Emergency clock reduction or shutdown
                throttle           System may request job termination

  THROTTLE DETECTION:
    nvidia-smi -q | grep -i throttle
    DCGM field: DCGM_FI_DEV_CLOCK_THROTTLE_REASONS
    Bitmask interpretation:
      0x0:  No throttle (target state)
      0x2:  Power limit throttle
      0x4:  Thermal throttle
      0x8:  Sync boost throttle
      0x10: Board limit throttle

  QUANTIFIED MFU IMPACT OF THERMAL THROTTLING:
    H100 base SM clock:  ~1.78 GHz
    H100 boost SM clock: ~1.98 GHz (full, no throttle)
    Throttled to base:   1.78 / 1.98 = 89.9% of peak -> 10.1% MFU loss

    At 10,000 GPUs with 10% throttle:
      Effective GPUs lost = 10,000 × 0.101 = 1,010 GPU-equivalents
      At $1.66/GPU-hr: 1,010 × 24 = 24,240 GPU-hrs/day = $40,238/day wasted
      Annual: $14.7M in wasted compute from thermal management failure alone

    At 5°C thermal violation (sustained 80°C vs 75°C target):
      Estimated frequency reduction: ~7%
      MFU impact: -7% across affected GPUs
      At 5,000 GPUs with 30% affected: 5,000 × 0.30 × 0.07 = 105 GPU-equivalents
lost
```

5.3 Cooling Technologies — Air, DLC, and Immersion

5.3.1 Air Cooling — The Baseline That Is Becoming Insufficient

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AIR COOLING CHARACTERISTICS:

  Architecture: Front-to-back airflow, cold aisle intake, hot aisle exhaust
  CRAC (Computer Room Air Conditioner) recirculates hot aisle air

  Performance parameters:
    GPU TDP supported:     Up to ~500W reliably (H100 PCIe = 350W feasible)
    H100 SXM (700W):       Marginal — requires high-velocity fans, hot aisle
containment
    B200 (1,000W):         Not viable for standard datacenter designs
    Effective rack density: Up to 30-40 kW/rack with optimized air

  Overhead costs:
    Fan power consumption: ~400-600W per DGX node (5-6% of server TDP)
    PUE range:             1.2-1.6 (typical modern facility: 1.25-1.35)

  WHY AIR IS INSUFFICIENT FOR B200-ERA AI:
    A full rack of 8 DGX B200 nodes: 8 nodes × ~14.3 kW = ~115 kW/rack
    Standard hot-aisle/cold-aisle designed for 10-30 kW/rack.
    Air velocity needed for 115 kW: creates acoustic and mechanical failures.
    Fan noise would exceed OSHA limits for worker proximity.
    Airflow-induced vibration damages storage and connectors.

  WHEN AIR COOLING IS STILL VALID:
    H100 PCIe inference nodes (350W, single GPU per standard server)
    A100-era training (400W, manageable in optimized DGX-A100 chassis)
    Retrofit scenarios where DLC cost exceeds 18-month power savings
    Development/lab environments with limited scale
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5.3.2 Direct Liquid Cooling (DLC) — The Current Production Standard

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DLC ARCHITECTURE (DGX H100 SuperPOD):

  Facility Water (20-30°C supply)
        |
        v
  Cooling Distribution Unit (CDU)
    - Heat exchanger (facility water <-> coolant loop)
    - Variable-speed pump
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    - Flow monitoring + leak detection
        |
        v
  Rack-level Manifold
        |
        |---- Server 1 Cold Plate Bars (one bar per GPU + CPU)
        |---- Server 2 Cold Plate Bars
        ...
        |---- Server 8 Cold Plate Bars

  Per-GPU cold plate specs (H100 SXM):
    Coolant inlet temperature:   ≤ 45°C  (facility-side)
    Coolant outlet temperature:  55-65°C (at full load)
    Thermal resistance:          ~0.02°C/W per GPU cold plate
    Junction-to-coolant ΔT:      ~14°C at 700W
    GPU junction temperature:    45°C + 14°C = 59°C  (well under 83°C limit)
    Heat removal fraction:       85-90% of server heat via DLC
                                 Residual 10-15% via residual air

DLC PERFORMANCE AND COST NUMBERS:

  Max GPU TDP supported:   1,500W+ (future B300 generation compatible)
  Rack density:            100-300 kW/rack
  PUE:                     1.03-1.15
  CDU CapEx per rack:      $50,000-200,000+ (amortized 5-7 years)
  Pump + sensor OpEx:      ~$100-200W overhead per rack (1-2% of TDP)
  Waste heat temperature:  55-65°C → usable for district heating
  Maintenance overhead:    Quarterly inspection of fittings and flow sensors

  DLC PUE SAVINGS vs AIR COOLING:
    Air (PUE 1.35) vs DLC (PUE 1.10) for 10,000 H100 cluster:
      Air power overhead:  7,000 kW IT × (1.35-1.0) = 2,450 kW cooling
      DLC power overhead:  7,000 kW IT × (1.10-1.0) = 700 kW cooling
      Savings:             1,750 kW in cooling power
    At $0.06/kWh:
      Annual savings:      1,750 kW × 8,760 hrs × $0.06 = $9.2M/year
    DLC installation cost: ~$25M (200 racks × $125K/rack)
    Payback period:        $25M / $9.2M = 2.7 years
    -> DLC pays for itself in under 3 years at H100-scale deployments.
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5.3.3 Immersion Cooling

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IMMERSION COOLING — BEYOND DLC:
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Servers submerged in dielectric fluid (engineered, non-conductive)
Heat transfers from all component surfaces to the fluid directly.

SINGLE-PHASE IMMERSION:
  Fluid remains liquid throughout.
  External heat exchanger removes heat; pump circulates fluid.
  Vendors: GRC (Green Revolution Cooling), LiquidStack
  Rack density: up to 400 kW/rack
  PUE: 1.02-1.08

TWO-PHASE IMMERSION:
  Fluid boils on hot components (latent heat transfer — very efficient).
  Vapor rises, condenses on chiller coil, drips back by gravity.
  Entirely passive fluid circulation.
  Vendors: 3M Novec (discontinued), LiquidStack BitCooler
  Rack density: up to 500 kW/rack
  PUE: 1.01-1.05

COOLING TECHNOLOGY COMPARISON TABLE:

	Air Cooling	DLC (Cold Plate)	Single-Phase Immersion	Two-Phase Immersion
Max GPU TDP	~500W	~1,500W	2,000W+	2,000W+
Rack density	≤40 kW	≤300 kW	≤400 kW	≤500 kW
PUE	1.2-1.6	1.03-1.15	1.02-1.08	1.01-1.05
Cooling CapEx/kW	$1-3K	$5-20K	$15-50K	$30-80K
Maintenance	Low	Medium	High	Very High
GPU serviceability	Easy	Moderate	Complex	Very Complex
Industry adoption	Widespread	Growing	Niche	Pilot phase
B200 viable?	No	Yes	Yes	Yes
5-year outlook	Declining	Dominant	Growing	R&D phase

REAR-DOOR HEAT EXCHANGERS (RDHx):
  Water-cooled door on rear of standard rack.
  Removes 60-80% of heat before exhaust enters hot aisle.
  Best fit: retrofit when upgrading to H100 in existing air-cooled facility.
  Rack density: 40-80 kW. PUE: 1.15-1.25.
  No server-side plumbing modifications required.

5.4.1 DVFS — The Core Power Control Mechanism

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DYNAMIC VOLTAGE AND FREQUENCY SCALING (DVFS):

  P_dynamic ∝ C × V² × f

  Goal: maximize f while keeping P ≤ TDP limit

  DVFS FEEDBACK LOOP (executes every ~1ms):
    1. Power sensor measures current draw
    2. Thermal sensor reads junction temperature
    3. DVFS controller evaluates:
       - Is current_power > power_limit?
       - Is T_j > thermal_threshold?
    4. If YES to either:
       a. Reduce V_core by one step (~5mV)
       b. Reduce f_core by one bin (~13 MHz)
       c. New P = C × (V-5mV)² × (f-13MHz) → lower power
    5. If NO to both AND headroom available:
       a. Increase V_core by one step
       b. Increase f_core by one bin
       c. This is "GPU boost" — auto-overclocking within TDP

  H100 FREQUENCY BINS:
    Max boost clock:  ~1,980 MHz (at thermal/power headroom)
    Base clock:       ~1,785 MHz (guaranteed at TDP, any temperature)
    Throttle floor:   ~1,350 MHz (emergency thermal throttle)
    Step size:        ~13 MHz per DVFS bin
    Steps from base to boost: (1980-1785)/13 ≈ 15 frequency steps
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5.4.2 GPU Power States (P-States)

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NVIDIA P-STATE REFERENCE (H100):

  P-State  Description          SM Clock    Power Draw    Use Case
  ──────────────────────────────────────────────────────────────────────
  P0       Maximum performance  1,785-1,980  650-700W     Active
inference/training
  P2       High performance     1,350-1,785  400-650W     Light workloads
  P4       Balanced             900-1,350    200-400W     Moderate utilization
  P8       Low performance      300-900      100-200W     Background tasks
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  P12      Minimum power        300          ~100W        Nearly idle
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  PERSISTENCE MODE for inference servers:
    By default, GPUs transition to P8/P12 between inference requests.
    This creates 100-400ms wake latency on first token of a new request.

    Fix: enable persistence mode
    nvidia-smi --persistence-mode=1

    With persistence mode: GPU stays in P0 between requests.
    Wake latency eliminated.
    Power increase: ~100-200W during idle periods.
    For production serving: almost always the correct tradeoff.

POWER MANAGEMENT COMMANDS:

  # Check current power draw and limits
  nvidia-smi -q | grep -A 10 "Power Readings"

  # Set power cap to 90% of TDP (630W for H100) for all GPUs
  sudo nvidia-smi -pl 630

  # Lock SM clocks to specific range (eliminates DVFS variance)
  sudo nvidia-smi -lgc 1785,1785  # lock to base clock for consistent benchmarks

  # Monitor real-time (1-second interval)
  nvidia-smi dmon -s pcut -d 1

  # DCGM power monitoring
  dcgmi dmon -e 155,156,157  # power usage, power limit, power state

THE 90-95% TDP RULE:
  Running at exactly 100% TDP triggers frequent DVFS steps.
  Frequency oscillates as GPU bounces between P0 and slight throttle.
  This creates latency variance — especially visible at P99 serving latency.

  At 90-95% TDP:
    Thermal headroom: comfortable margin above throttle threshold
    Performance: ~98-99% of full boost (frequency is barely reduced)
    Latency variance: eliminated (no DVFS bouncing)
    Production recommendation: set power cap to 90-95% TDP for all
    serving workloads where consistent P99 latency matters.
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5.5 Power Budgeting — From Chip to Rack to Datacenter

5.5.1 Rack Power Density Evolution

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RACK POWER DENSITY BY GPU GENERATION:

  GPU Generation  TDP    DGX Node  8-Node Rack  Notes
  ──────────────────────────────────────────────────────────────
  V100 (2018)     300W   2.4 kW    19 kW        Standard air cooling viable
  A100 (2020)     400W   6.5 kW    52 kW        High-velocity air, marginal
  H100 (2022)     700W   10.2 kW   82 kW        DLC strongly recommended
  B200 (2024)    1000W   14.3 kW   115 kW       DLC mandatory
  B300/GB300     1200W+  ~18 kW   ~144 kW       DLC + high-temp coolant req.
  ──────────────────────────────────────────────────────────────
  Legacy DC floor: 5-10 kW/rack  (completely inadequate for AI)
  Modern enterprise: 15-30 kW/rack (marginal for H100)
  AI-purpose-built: 40-120+ kW/rack (required for H100/B200)

COMPLETE RACK POWER BUDGET (8× H100 SXM5 DGX node):

  Component                     Power      Notes
  ──────────────────────────────────────────────────────────────
  8× H100 GPU                  5,600W    8 × 700W
  2× CPU (Xeon/EPYC)             400W    2 × 200W typical
  DRAM (DDR5, 2 TB)              200W    ~100W/TB
  NVMe storage                    80W    4 × 2 TB NVMe
  InfiniBand HCA (2× 400G)       100W    2 × 50W
  Fans (even in DLC node)        200W    Residual air cooling
  PSU overhead (3% efficiency)   200W    3% loss at 85% load
  Networking switches (shared)    50W    Pro-rated per node
  DLC pump + sensors              25W    CDU overhead share
  ──────────────────────────────────────────────────────────────
  TOTAL PER NODE                 6,855W   ~6.9 kW per node
  TOTAL PER 8-NODE RACK         54,840W   ~55 kW per rack
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5.5.2 Power Usage Effectiveness (PUE)

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PUE DEFINITION AND CALCULATION:

  PUE = Total Facility Power / IT Equipment Power
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  MEANING: A PUE of 1.3 means that for every 1W of useful compute,
  0.3W is spent on overhead (cooling, UPS losses, lighting, etc.).

WORKED EXAMPLE — 10,000 GPU H100 CLUSTER:

  IT Equipment Power:
    10,000 × H100 GPUs @ 700W:           7,000 kW
    CPU servers (1:8 ratio):               313 kW
    InfiniBand switches:                   150 kW
    Storage (Lustre/GPFS):                 200 kW
    Total IT Power:                      7,663 kW

  Overhead Power (DLC datacenter):
    DLC cooling system:                    690 kW  (9%)
    UPS losses:                            307 kW  (4%)
    Lighting, security, misc:               30 kW  (<0.5%)
    Total Overhead:                      1,027 kW

  PUE = (7,663 + 1,027) / 7,663 = 1.134

PUE BENCHMARKS:
  PUE = 1.00:   Theoretical perfect (physically impossible)
  PUE = 1.02:   World-class (Google Oregon with waste heat reuse)
  PUE = 1.05-1.15: Excellent (modern hyperscaler with DLC)
  PUE = 1.2-1.3:   Good (modern air-cooled facility)
  PUE = 1.4-1.6:   Average (typical enterprise datacenter)
  PUE > 2.0:    Legacy (half of IT power wasted on cooling)

PUE IMPACT ON ANNUAL POWER COST (10,000 H100 cluster, $0.06/kWh):

  IT Power only:   7,663 kW × 8,760 hrs × $0.06 = $40.3M/year
  PUE 1.10:        7,663 × 1.10 × 8,760 × $0.06 = $44.3M/year  (+$4.0M)
  PUE 1.30:        7,663 × 1.30 × 8,760 × $0.06 = $52.4M/year  (+$12.1M)
  PUE 1.50:        7,663 × 1.50 × 8,760 × $0.06 = $60.4M/year  (+$20.1M)

  Every 0.10 improvement in PUE saves ~$4M/year on this cluster.
  A $25M DLC installation that improves PUE from 1.35 → 1.10 saves $10.5M/year.
  Payback: 2.4 years.
```

5.6 TCO Modeling for AI Infrastructure

5.6.1 TCO Framework — CapEx vs. OpEx

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TCO BREAKDOWN — 1,024 H100 SXM5 CLUSTER (3-YEAR HORIZON): [ESTIMATED]

CAPEX (one-time, amortized over 3 years):
  Component                    Cost
  ──────────────────────────────────────────────────────────────
  GPU hardware (H100 SXM5)     1,024 × ~$30K     = $30.7M
  CPU servers + chassis         128 DGX nodes     = $5.1M
  InfiniBand HDR2 fabric        ~512 ports        = $3.2M
  Storage (Lustre)              ~2 PB             = $1.5M
  DLC cooling infrastructure    128 racks × CDU   = $6.4M
  Datacenter buildout           power + facility  = $2.0M
  ──────────────────────────────────────────────────────────────
  TOTAL CAPEX                                     = ~$48.9M
  Amortized per year (3-yr)                       = ~$16.3M/year

OPEX (recurring annual costs):
  Component                    Annual Cost
  ──────────────────────────────────────────────────────────────
  Power (1,024 × 700W × PUE 1.10 × 8,760h × $0.06)  = $4.0M/year
  DLC maintenance + coolant                           = $0.6M/year
  Operations staff (~5 FTEs × $180K)                 = $0.9M/year
  Hardware maintenance (2% of CapEx)                 = $1.0M/year
  Software licenses (SLURM, monitoring)              = $0.3M/year
  ──────────────────────────────────────────────────────────────
  TOTAL ANNUAL OPEX                                  = $6.8M/year

3-YEAR TCO:
  CapEx: $48.9M + OpEx: 3 × $6.8M = $48.9M + $20.4M = $69.3M

COST PER GPU-HOUR (3 years, 80% utilization):
  Total GPU-hours = 1,024 GPUs × 8,760 hrs × 3 yrs × 0.80 = 21.6M GPU-hours
  Cost/GPU-hour = $69.3M / 21.6M = $3.21/GPU-hour (owned infra, DLC)

  Cloud H100 equivalent:
    On-demand (AWS p4de): ~$8-12/GPU-hour
    1-year reserved:      ~$3-4/GPU-hour
    3-year reserved:      ~$2-2.50/GPU-hour
```

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  BREAKEVEN: owned hardware competes with 3-year cloud reserved at ~70%+
utilization.
```

5.6.2 Cost Per Token — Connecting TCO to LLM Business Metrics

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POWER COST PER TOKEN (LLaMA-3 70B inference, H100):

  Given:
    Throughput at batch=32:   ~2,000 tokens/sec per GPU (decode, FP8)
    GPU power draw:           700W
    Electricity:              $0.05/kWh
    PUE:                      1.10

  Power cost per 1M tokens:
    Time for 1M tokens:       1,000,000 / 2,000 = 500 seconds
    Energy:                   700W × 1.10 × (500/3600) hrs = 0.107 kWh
    Power cost:               0.107 × $0.05 = $0.0053 per 1M tokens

OPTIMIZATION IMPACT ON COST PER TOKEN:

  Configuration          Throughput   Time/1M tok  Energy/1M  Cost/1M
  ──────────────────────────────────────────────────────────────────────
  BF16 baseline          1,200 tok/s  833 sec      0.178 kWh  $0.0089
  FP8 quantization       2,000 tok/s  500 sec      0.107 kWh  $0.0054  -40%
  INT4 quantization      3,500 tok/s  286 sec      0.061 kWh  $0.0031  -66%
  INT4 + power cap 560W  3,500 tok/s  286 sec      0.048 kWh  $0.0024  -73%
  INT4 + FA3             4,200 tok/s  238 sec      0.040 kWh  $0.0020  -78%
  ──────────────────────────────────────────────────────────────────────

  AT 100B TOKENS/MONTH (ChatGPT-scale):
    BF16 baseline:   100B × $0.0089/1M = $890/month per GPU in power
    Fully optimized: 100B × $0.0020/1M = $200/month per GPU in power
    Savings per GPU: $690/month
    At 10,000 GPUs:  $6.9M/month saved = $82.8M/year

  This is the business case for every optimization in Parts III-V of this book.
  Performance optimization IS cost optimization at scale.
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5.7 Performance-Per-Watt — The True Optimization Target

Raw TFLOPS or tokens/sec is the wrong optimization metric for production AI systems.

Performance-per-watt captures the real constraint: cost and carbon, not just speed.

```
PERFORMANCE-PER-WATT COMPARISON (LLaMA-3 70B FP8 inference):

  GPU          TDP     Tok/s (FP8)  Tok/s/W    vs A100   Notes
  ──────────────────────────────────────────────────────────────
  A100 SXM4    400W    1,100        2.75       1.0×      Baseline
  H100 SXM5    700W    2,200        3.14       1.14×     +14% efficiency
  H200 SXM5    700W    3,100        4.43       1.61×     +61% (same TDP!)
  MI300X       750W    2,800        3.73       1.36×     single-GPU 70B
  B200 SXM    1000W    6,800        6.80       2.47×     best efficiency
  ──────────────────────────────────────────────────────────────

  KEY INSIGHT:
    H200's 41% efficiency gain over H100 at IDENTICAL TDP (700W).
    This means: for a fixed power budget, H200 serves 41% more tokens.
    If your datacenter power is the constraint (not hardware cost),
    H200 is strictly superior to H100 regardless of price premium.

    B200's 2.47× efficiency at 1.43× TDP (1000 vs 700W):
    B200 does 2.47× the work of A100 while consuming 2.5× the power.
    Performance-per-watt is nearly identical to A100 in absolute terms —
    but the TOTAL throughput per watt-budget is 2.47× better.
    This is why B200 racks need more power but fewer racks for same output.

DVFS EFFICIENCY CURVE:
  Clock %   Power     Throughput   Efficiency (tok/s/W)  vs 100%
  ──────────────────────────────────────────────────────────────
  100%      700W      2,200 tok/s  3.14                  baseline
  95%       630W      2,160 tok/s  3.43                  +9.2%
  90%       560W      2,090 tok/s  3.73                  +18.8%
  80%       448W      1,980 tok/s  4.42                  +40.8%
  70%       350W      1,800 tok/s  5.14                  +63.7%
  ──────────────────────────────────────────────────────────────

  The efficiency sweet spot for inference: 80-90% of max clock.
  Running at 90% clock: 5% throughput loss, 20% power reduction, 19% efficiency
gain.
  If the cluster is power-constrained (rack density limited):
    Drop to 90% clock, pack 20% more GPUs per rack, net +15% throughput.
  If the cluster is throughput-constrained (need max tokens/sec):
    Run at full boost, accept higher power draw.
```

5.8 Build vs. Buy vs. Rent — The Decision Framework

```
AI COMPUTE PROCUREMENT DECISION FRAMEWORK:

STEP 1 — UTILIZATION RATE (most important variable):
  Utilization < 30%?        -> RENT (cloud on-demand)
  Utilization 30-60%?       -> HYBRID (reserved cloud + spot)
  Utilization > 60%?        -> BUY (on-premise or colo)
  Utilization > 80%?        -> DEFINITELY OWN (cloud becomes 2-3× more expensive)

STEP 2 — TIMELINE:
  Need in < 3 months?       -> RENT (GPU hardware lead time: 6-18 months)
  Need in > 6 months?       -> BUY viable (plan procurement now)
  Need for < 18 months?     -> RENT (3-year amortization doesn't work)
  Need for > 2 years?       -> BUY (amortization economics favorable)

STEP 3 — BREAKEVEN CALCULATION:
  Cloud H100 (3-year reserved):  ~$2.50/GPU-hour
  On-prem H100 (owned, 80% util): ~$3.21/GPU-hour (from §5.6.1)
  On-prem at higher utilization:
    At 90% util: $69.3M / (1,024 × 8760 × 3 × 0.90) = $2.85/GPU-hour
    At 95% util: $69.3M / (1,024 × 8760 × 3 × 0.95) = $2.70/GPU-hour

  BREAKEVEN UTILIZATION (owned vs 3-year reserved cloud):
    ~70-75% average utilization over 3 years

  Practical breakeven is NOT a fixed number — it depends on:
    GPU acquisition price (H100 prices dropped ~40% since launch)
    Electricity cost (varies by region: $0.02-0.15/kWh)
    Depreciation schedule (3 vs 4 years changes the math significantly)
    Operational overhead (on-prem needs dedicated 5-10 FTE)

STEP 4 — ADDITIONAL RISK FACTORS:
  Technology obsolescence:  New GPU generation every 2 years.
                            2024 H100 purchase may be "old" by 2026.
                            Cloud stays current; owned hardware depreciates faster
                            in a GPU-generation-every-2-years world.

  Power infrastructure:     Legacy facilities may need $500K-5M retrofits.
                            Add to CapEx before comparing to cloud.

  Operational burden:       On-prem requires 24/7 ops, hardware replacement,
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                              driver management, thermal incidents.
                              Estimate 5-10 FTEs for 1,024 GPU cluster.

  Network egress:             Cloud training produces model artifacts.
                              Large model → large egress fees to bring data home.

DECISION MATRIX:

  Scenario                                Recommendation    Rationale
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  Stable training, >70% util, 3+ years Own (colo/DC)     2× cheaper long-term
  Bursty/seasonal inference               Cloud + auto-scale Avoid idle CapEx
  <12 months of workload                  Cloud             Amortization doesn't
work
  Regulatory data residency required      Own               Cloud may not comply
  Frontier model training (10K+ GPUs)     Own + Cloud burst CapEx + on-demand
  Research/experimental                   Cloud spot        Tolerate preemption
  Startup (<18 months horizon)            Cloud             Risk and lead time
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5.9 Thermal Monitoring and Remediation

5.9.1 The Three-Layer Monitoring Stack

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THERMAL MONITORING ARCHITECTURE:

LAYER 1 — SILICON (per GPU, per second):
  Source: DCGM (NVIDIA DCGM daemon) + nvidia-smi

  Critical metrics and alert thresholds:
    GPU_TEMP (T_j):     > 78°C = Warning; > 82°C = Critical
    MEM_TEMP (HBM):     > 85°C = Warning; > 90°C = Critical (HBM max ~95°C)
    POWER_DRAW:         > 95% of power cap = Warning (approaching throttle)
    SM_CLOCK:           < 1,785 MHz during active workload = Throttle detected
    CLOCK_THROTTLE:     Any non-zero = Immediate investigation
    PERF_STATE:         P2 or higher during active workload = Under-powered

  DCGM monitoring commands:
    dcgmi dmon -e 100,150,155,156,1001,1002  # T, power, clocks
    dcgmi health -g 0 -j                     # JSON health summary
```

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LAYER 2 — SERVER / NODE (per server, per 10 seconds):
  Source: IPMI/BMC (Baseboard Management Controller)
  Metrics:
    Inlet air temperature (ASHRAE sensor, server front)
    Fan RPM and redundancy status
    PSU input/output power and efficiency
    VRM temperature for CPU and GPU voltage regulators
    Coolant flow rate + inlet/outlet temperature delta (DLC)

LAYER 3 — FACILITY (per rack/row, per minute):
  Source: DCIM software + CDU instrumentation
  Metrics:
    Rack-level power draw (PDU metering)
    CDU coolant temperature and flow rate
    Hot/cold aisle temperature gradient
    PUE (calculated real-time)
    UPS load and battery state of charge
```

5.9.2 Thermal Incident Response Protocol

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THERMAL INCIDENT RESPONSE — TIERED ESCALATION:

TIER 1: T_j 75-80°C (Warning — soft throttle zone)
  Action: Investigate root cause within 1 hour
  Diagnosis checklist:
    1. Check if workload has changed (new training job with larger batch?)
    2. Check CDU/facility temperatures (is cooling system degraded?)
    3. Check DLC flow rate (is pump delivering adequate flow?)
    4. Check ambient: is hot aisle recirculating into cold aisle intake?
  Mitigation: Reduce GPU power cap by 5-10% to stabilize temperature

TIER 2: T_j 80-83°C (Critical — hard throttle warning)
  Action: Immediate intervention
  Mitigation options (in order of speed):
    a. nvidia-smi -pl [lower_value]  (reduce power cap in minutes)
    b. Reduce batch size / request rate  (reduces GPU activity factor)
    c. Engage emergency cooling (increase CDU pump speed, check chiller)
  Escalation: page facilities team if not resolved in 30 minutes

TIER 3: T_j > 83°C (Emergency — hardware protection engaged)
  Action: GPU has activated hardware protection.
          Performance penalty is severe (>15% throughput loss).
  Response: Reduce load or throttle the specific node.
```

```
  Diagnostic:
    nvidia-smi -q | grep -i thermal
    Check for DCGM_FI_DEV_CLOCK_THROTTLE_REASONS bitmask = 0x4

HBM TEMPERATURE MONITORING (CRITICAL — often overlooked):
  HBM has a lower thermal limit than the GPU die.
  HBM temperature exceeding limit causes immediate GPU shutdown.
  Monitor: DCGM_FI_DEV_MEM_COPY_THROTTLE_ACTIVE

  nvidia-smi queries for HBM temperature (H100):
    nvidia-smi -q | grep "Memory Current Temp"
  Alert at > 85°C. Hard limit at ~95°C (driver forces shutdown).
```

5.9.3 Thermal Best Practices for Production Clusters

```
THERMAL ENGINEERING BEST PRACTICES:

  1. SET POWER CAPS PROACTIVELY:
     Default: 90-95% of TDP for all production serving workloads
     Training: 95-100% TDP acceptable if thermal monitoring is in place
     Command: sudo nvidia-smi -pl [value] -i [GPU_index]
     Persistent across reboots:
       Edit /etc/systemd/system/nvidia-powerlimit.service

  2. ENABLE PERSISTENCE MODE:
     sudo nvidia-smi --persistence-mode=1
     Prevents P-state cycling between inference requests.
     Eliminates 100-400ms latency spikes from GPU wake transitions.

  3. ASYMMETRIC WORKLOAD PLACEMENT:
     LLM decode (low power, ~400-500W) vs. training (high power, ~650-700W).
     Mix workloads across racks to avoid power hotspots.
     Thermal-aware scheduling: prefer cooler nodes for new training jobs.

  4. VALIDATE DLC FLOW RATES QUARTERLY:
     Degraded pump or blocked filter → reduced flow → thermal violations.
     Target: inlet ≤ 45°C, outlet ≤ 65°C, ΔT ≤ 20°C at full load.

  5. ECC — NEVER DISABLE FOR PERFORMANCE:
     H100 HBM ECC adds ~3-5% memory bandwidth overhead.
     For memory-bandwidth-bound decode: this is a real ~3% throughput cost.
     Disabling ECC: enables Silent Data Corruption events.
     Silent Data Corruption corrupts model weights mid-training silently.
```

```
    A 3% performance improvement that risks corrupting a 30-day training
    run is never worth it. ECC must remain enabled in all production GPUs.
```

5.10 Key Takeaways and Review Questions

```
THERMAL AND POWER NUMBERS TO KNOW COLD:

  GPU TDP:
    H100 SXM5:    700W  (SXM, DLC recommended)
    H200 SXM5:    700W  (same TDP, 43% more inference throughput)
    B200 SXM:    1000W  (DLC mandatory)
    MI300X (AMD): 750W

  H100 Thermal Limits:
    Max operational T_j:  83°C
    Soft throttle onset:  ~75°C (~1-3% clock reduction per °C above this)
    HBM temperature max:  ~95°C
    Healthy operational:  ≤ 70°C at sustained load

  Cooling PUE Reference:
    Air cooling:      1.2-1.6
    Rear-door HX:     1.15-1.25
    Direct Liquid:    1.03-1.15  (target for new AI deployments)
    Immersion:        1.01-1.08  (emerging, high maintenance)

  Power Cost (10K H100 cluster, $0.05/kWh, PUE 1.10):
    Annual power cost: ~$34M/year
    Per 0.10 PUE improvement: ~$4M/year saved

  Performance-Per-Watt (LLaMA-3 70B FP8):
    A100:   2.75 tok/s/W  (baseline)
    H100:   3.14 tok/s/W  (+14%)
    H200:   4.43 tok/s/W  (+61%)
    B200:   6.80 tok/s/W  (+147%)

KEY PRINCIPLES (r3 integration):

  1. THERMAL THROTTLE IS DIRECT MFU LOSS:
     5°C above target = 7% frequency reduction = 700 GPU-equivalents lost
     at 10,000 GPUs. Monitor T_j every second. React before throttle engages.
```

```
  2. DYNAMIC POWER SCALES AS V^2 — USE IT:
     10% voltage reduction → 19% power reduction → 10% frequency reduction.
     The last 10% of frequency costs 20%+ of power.
     90-95% TDP is the efficiency optimum for serving.

  3. DLC IS NOT OPTIONAL ABOVE 700W TDP:
     Air cooling cannot economically remove 1,000W+ per GPU.
     DLC PUE (1.05-1.15) vs air (1.2-1.6) = $M/year operating cost difference.
     $25M CDU investment recouped in 2-3 years at H100 scale.

  4. POWER COST IS CURRENTLY 3-4% OF TCO BUT GROWING:
     At today's GPU prices, hardware amortization dominates.
     As GPU prices halve each generation, power becomes 10-15% of TCO.
     Engineers who optimize for tok/s/W today are ahead of the curve.

  5. TCO BREAKEVEN IS ~70% UTILIZATION AT 18 MONTHS:
     Below this: cloud is cheaper. Above it: owned hardware wins on 3-year NPV.
     The breakeven is not fixed — recalculate with current GPU pricing.

  6. PERFORMANCE-PER-WATT IS THE CORRECT TARGET, NOT RAW TFLOPS:
     H200 = H100 at 41% more efficiency at same TDP.
     B200 = 2.47× A100 efficiency at 2.5× TDP.
     Procurement based on TFLOPS/$ without TFLOPS/W over-invests in
infrastructure.

  7. NEVER DISABLE ECC:
     3-5% bandwidth overhead is real on memory-bound workloads.
     The alternative — silent weight corruption mid-training — is catastrophic.
```

Review Questions

Difficulty: Intermediate

Question 1 (Thermal Throttle Quantification): A 5,000-GPU H100 SXM5 training cluster reports average junction temperatures of 80°C (vs the 75°C target) across 30% of GPUs during peak training hours. The cluster is in an air-cooled datacenter achieving PUE 1.35. (a) Estimate the MFU impact on the throttling GPUs. (b) Calculate the daily and annual GPU-hour cost at $1.66/GPU-hour fully loaded. (c) Provide three specific engineering interventions ordered by implementation speed.

Expected answer: (a) At 80°C (5°C above 75°C soft throttle threshold), estimated frequency reduction ≈ 5×3% = ~15% → but more conservatively, H100 hard throttle is at 83°C. At 80°C in hard throttle zone: frequency reduction ~7-10%. MFU impact: ~7% loss on the 30% affected GPUs. Effective lost compute = 5,000 × 0.30 × 0.07 = 105 GPU-equivalents constantly throttled.

(b) Daily: 105 GPU-equivalents × 24 hours × $1.66 = $4,183/day. Annual: $4,183 × 365 = $1.53M/year wasted from thermal management failure. (c) Fastest: (1) Reduce GPU power cap to 90% TDP (630W) via `sudo nvidia-smi -pl 630` — implementable in minutes, recovers ~3-5°C immediately with ~5% throughput cost. (2) Increase CDU coolant flow rate and verify inlet temperature ≤ 40°C — if DLC present; if air-cooled, add temporary portable cooling units to affected rows. (3) Thermal-aware workload scheduling: migrate training jobs from hottest 30% of nodes to cooler nodes; reduce per-node batch size on affected GPUs. Long-term: replace air cooling with DLC — eliminates the problem entirely.

Difficulty: Principal Level

Question 2 (Cooling Technology Selection): You are advising on datacenter buildout for 10,000 B200 GPUs (1,000W TDP). The facilities team proposes upgrading the existing air-cooled datacenter with new high-velocity fans and containment. Make the quantitative case for DLC instead. Include: PUE comparison, annual power cost difference at $0.06/kWh, DLC CapEx, and payback period. Under what conditions is the air-cooling proposal defensible?

Expected answer: Air cooling at 1,000W per GPU: Δ T_j with air thermal resistance 0.07°C/W = 1000 × 0.07 = 70°C rise above ambient. Required ambient ≤ 83 - 70 = 13°C — physically impossible in a real datacenter. Air cooling is not viable at B200 TDP regardless of fan upgrades. This is a physics argument, not an economics argument. PUE comparison: DLC (1.10) vs. theoretical air upgrade (1.25). IT power: 10,000 × 1,000W = 10 MW. Annual power cost: DLC = 10 MW × 1.10 × 8,760 × $0.06 = $57.8M. Air = 10 MW × 1.25 × 8,760 × $0.06 = $65.7M. Annual savings from DLC: $7.9M. DLC CapEx: 10,000 B200 / 8 per rack × 125 racks × $125K/rack CDU = $15.6M (CDUs alone, not including piping). Payback: $15.6M / $7.9M = 1.97 years. Air cooling is defensible only if: (a) the cluster will be retired in <18 months before DLC pays back, (b) facility renovation for DLC piping costs exceed $30M+ making payback >4 years, or (c) immersion cooling is being considered instead of DLC, making DLC an intermediate step not worth the investment.

Difficulty: Principal Level

Question 3 (TCO Model Construction): Build a 3-year TCO comparison for deploying LLaMA-3 70B serving at 50,000 RPM, comparing: (A) 64 owned H100 SXM5 GPUs in a colocation facility vs. (B) equivalent cloud capacity on a 3-year reserved contract. State clearly which option you recommend and under what constraints the recommendation inverts.

Expected answer: Capacity needed: 50,000 RPM × assume 1,000 tokens/response = 50M tokens/sec (no), actually 50,000 requests/minute ÷ 60 = 833 RPS. Each H100 at batch=32 serving ≈ 2,000 tok/s decode. Average 500 tokens/response = each request takes 250ms. At 833 RPS: need 833 × 0.25s = ~208 concurrent requests. At 32 batch/GPU: 208/32 = ~7 GPUs needed for

throughput. For reliability (2× headroom): 14 GPUs. With 70B model needing TP=2: round to 16 GPUs (8 serving instances). For 24/7 availability with maintenance headroom: 32 GPUs minimum; 64 GPUs for full N+N redundancy. Option A (owned, 64 H100 SXM5): CapEx: 64 × $30K = $1.92M + servers/network $0.5M = $2.42M. Power (PUE 1.15, colo): 64 × 0.7kW × 1.15 × 8760 × $0.10/kWh (colo rate) = $450K/year. Colo hosting: $2,000/rack/month × 4 racks = $96K/year. Staff: 0.5 FTE = $90K/year. 3-year total: $2.42M + 3 × $636K = $4.33M. Option B (cloud, 3-year reserved): 64 H100-equiv × $2.50/GPU-hr × 8,760 × 3 = $4.22M. Result: Cloud is slightly cheaper ($4.22M vs $4.33M) primarily because 64 GPUs is too small a deployment for owned hardware to win on economics. Break-even: at ~70% utilization and > ~256 GPU scale, owned wins. Recommendation inverts if: (a) utilization drops below 60% (cloud becomes clearly better), (b) GPU price declines further before 3-year contract ends (owned hardware depreciates in capability), (c) workload duration < 18 months (amortization doesn't work), (d) team lacks ops expertise for GPU cluster management.

Difficulty: Principal Level

Question 4 (DVFS and Power Budgeting): Your datacenter has 12 MW of available IT power. A proposal deploys 1,200 B200 SXM GPUs (1,000W TDP each) plus supporting infrastructure. (a) Verify whether the power budget is feasible. (b) If infeasible, what is the maximum B200 count within budget? (c) The facilities team offers a 10% power overage during peak training. Explain why this creates more risk than it resolves.

Expected answer: (a) Full system power estimate: 1,200 × 1,000W = 1,200 kW GPU. Supporting infrastructure: CPUs (150 GPUs) = 1 CPU-server per 8 GPUs = 150 nodes × 350W = 52.5 kW. InfiniBand switches = 75 kW. Storage = 50 kW. Networking/misc = 40 kW. Total IT = 1,200 + 52.5 + 75 + 50 + 40 = 1,417.5 kW = 1.42 MW IT. At PUE 1.10: facility power = 1.42 × 1.10 = 1.56 MW. This is well under 12 MW — feasible! (Unless the question intends a much larger cluster or 12 MW is the entire facility with other tenants.) Revised interpretation for a meaningful problem: assume 12 MW total facility with DLC PUE 1.10 and other IT loads consuming 4 MW: available IT for GPUs = (12/1.10) - 4 = 6.9 MW for new GPUs. At 1,000W/GPU: 6,900 GPUs maximum. 1,200 GPUs is feasible. (c) Power overage during training creates three risks: (1) UPS bypass: most UPS systems are sized exactly at facility power + 5% margin. A 10% overage bypasses UPS protection for some circuits — a power fault during the overage period means unprotected equipment. A training run corrupted by power loss at step 25,000 of 30,000 wastes weeks of compute. (2) Thermal cascade: B200 at 1,100W (10% overage) during sustained training pushes T_j significantly above design point. DLC cooling is sized for 1,000W — at 1,100W, coolant return temperature rises, CDU struggles to maintain setpoint, thermal cascade begins. (3) HVAC suppression makes it worse: reducing HVAC to "make room" for GPU overage removes cooling precisely when more cooling is needed. The correct response: accept the power constraint, size the cluster within budget, and use FP8 training to improve throughput-per-watt rather than overclocking.

Further Reading

1. *ASHRAE TC 9.9 Thermal Guidelines for Data Processing Environments* — Fifth Edition (2021). The industry standard for thermal design. Defines class A, B, C inlet air temperatures. `ashrae.org`
2. *Google Sustainability Report — Data Center Energy* (annual). The most detailed public disclosure of hyperscaler PUE and efficiency practices. `sustainability.google/reports`
3. *Uptime Institute Global Data Center Survey* (annual). Industry-wide PUE benchmarks and cooling technology adoption trends. `uptimeinstitute.com`
4. *NVIDIA Datacenter Design Guide for DGX SuperPOD* (2023). The definitive reference for DLC design with H100. Coolant specs, flow rates, rack design. `docs.nvidia.com/ai-enterprise/dgx-superpod`
5. *Dynamic Voltage and Frequency Scaling for AI Accelerators* — See NVIDIA GTC 2023: "Power Efficiency in Large-Scale AI Training." The most accessible treatment of DVFS behavior in production H100 deployments.
6. *The Cloud vs. On-Premises AI Infrastructure Decision* — MLCommons (2024). Quantitative framework for build vs. buy analysis at various GPU scales. `mlcommons.org`

— End of Chapter 5 —

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